Marie Hoffmann, PhD

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01 PROFESSIONAL SUMMARY

Dr. Hoffmann is a Ph.D. computational statistician with more than a decade of experience building, validating, and independently evaluating predictive and causal models. Her work combines enterprise marketing-measurement leadership — marketing mix modeling, attribution, incrementality, and lift — with peer-reviewed research in uncertainty quantification, robustness, and high-dimensional model selection. Early-career audit and Sarbanes-Oxley experience grounds her orientation toward evidence, reproducibility, and independence.

02 AREAS OF EXPERTISE

Marketing Measurement & Incrementality. Marketing mix modeling (MMM), multi-touch attribution (MTA), incrementality and lift measurement, geo experimentation, market-response and price-elasticity modeling, and media ROI analysis.

Causal Inference & Experimental Validity. Randomized and quasi-experimental design, difference-in-differences, geo-lift and A/B testing, treatment-versus-control validity, confounding and alternative explanations, and the distinction between causal and associational claims.

Statistical Reliability, Uncertainty & Robustness. Prediction intervals and uncertainty quantification, calibration and delivered-versus-stated coverage, robustness under contamination and anomalous observations, heterogeneous-population and subgroup structure, high-dimensional variable selection, and sensitivity and stability analysis.

Model Validation, Reproducibility & Methodology Review. Model specification and diagnostics, independent vendor-model validation, reproducibility and documentation, benchmarking and cross-validation, transformation and implementation review, simulation-based stress testing, and data and instrumentation integrity.

03 PROFESSIONAL EXPERIENCE

Marketing Data Science Manager — MAY 2026 – PRESENT

University of Phoenix

- Leads the marketing mix modeling workstream with full ownership of MMM development and incremental-impact optimization — development, validation, simulation, and optimization — and owns executive alignment, VP-level stakeholder management, and methodology decisions for the MMM practice.
- Applies causal inference and experimentation (geo-lift, incrementality, A/B) across multi-stage conversion funnels; builds calibrated rare-event propensity models with explicit invalid-traffic detection, calibration, and temporal-drift diagnostics; and translates model logic into production decision rules.

Senior Data Scientist, Measurement Science — NOV 2025 – FEB 2026

Southern Glazer's Wine & Spirits

- Led executive KPI governance and measurement architecture — metric definitions, engagement-to-outcome KPI ladders, and documentation — for a new digital sales tool.
- Identified instrumentation gaps limiting reliable measurement, and surfaced and drove remediation of a cross-system measurement-inconsistency risk toward a unified system of record.

- Distinguished what the existing data could and could not yet support, and maintained analytic continuity through a coverage gap.

Principal Data Scientist — Marketing Measurement Lead

MAY 2020 – DEC 2024

HP Inc.

- Advanced from Lead Data Scientist, Pricing & Advanced Analytics (May 2020 – Jan 2023) to Principal Data Scientist and worldwide Marketing Measurement Lead (Jan 2023 – Dec 2024).
- **Methodology architecture.** Designed HP's initial in-house MMM/MROI and incrementality workflow, authored the end-to-end methodology, and defined the framework for how MMM, multi-touch attribution, and experimentation should be used at different decision levels.
- **Hands-on implementation.** Developed the core data preparation, model building, simulation, validation, optimization, and reproducibility tooling — the basis both for large throughput gains and for the ability to audit a measurement stack at the implementation level.
- **Independent model audit.** Instituted a standardized vendor-model validation and audit standard — requiring model equations, coefficient and uncertainty statistics, fit and cross-validation metrics, transformation documentation, and reproducible incremental-impact outputs — negotiated into vendor agreements so third-party models could be independently validated.
- **Vendor-neutral evaluation.** Led technical evaluation and validation across multiple external measurement methodologies and providers, and across 20+ deployed models.
- **Methodology communication & training.** Authored and delivered curricula in econometric modeling, causal inference applied to marketing, measurement frameworks, and diagnostics and validation.

R&D Data Scientist — Personal Systems Quality

JAN 2025 – NOV 2025

HP Inc.

- Applied large-scale predictive modeling to hardware reliability — rare-event / anomaly detection of component failures across high-volume device telemetry, with target event rates below 1% across billions of records and millions of individual serial numbers.
- Built a tunable machine-learning detection model series with explicit control of the false-positive / true-positive trade-off.
- Developed full-life telemetry trend detection and reusable cumulative-stress-indicator workflows, with data-treatment, validation, and visualization tooling to confirm data accuracy and model performance.

Senior Data Scientist

NOV 2019 – JUN 2020

Solsten

- Behavioral and psychometric segmentation linking user traits to marketing tactics; quantified personalization impact through causal modeling and adaptive experimentation.

Lead Data Scientist

FEB 2018 – NOV 2019

Age of Learning

- Led the analysis of a randomized controlled efficacy study of an adaptive early-mathematics product (886 students; 452 treatment, 434 control; two school districts), estimating causal learning gains with hierarchical linear (mixed-effects) models controlling for grade, district, teacher, demographics, and prior knowledge; conducted power analysis and simulation and IRT psychometric analysis of the assessments (Cronbach's $\alpha \approx 0.9$).
- Designed and deployed a live struggle-detection (wheel-spinning) model: built a behavior-coding framework (2,395 students coded; inter-rater reliability Cohen's $\kappa = 0.71$); compared classifiers with 10-fold cross-validation and class resampling; applied nonparametric prediction intervals from her doctoral research to produce per-student uncertainty estimates; and validated across product versions, with an early-detection variant predicting later-stage struggle from early behavior.
- Led methodology governance and technical training for the data-science team (including a full random-forests curriculum) and set scientific-rigor standards across the team's modeling work.

04 EARLIER EXPERIENCE — AUDIT & FINANCIAL CONTROLS

Early-career internships providing formal grounding in assurance, internal controls, documentation, and evidentiary review.

Audit & Assurance Intern (Public Sector) — Deloitte

SUMMER 2007

Public-sector external financial audit and assurance: financial-record and transaction testing, internal-controls evaluation, and working-paper preparation.

Sarbanes-Oxley Project Team — Rio Tinto Alcan

SUMMER 2006

SOX internal-controls testing over financial reporting — design and operating-effectiveness assessment, evidence collection, and control-deficiency documentation — across US and Mexico locations, conducted in English and Spanish.

05 RESEARCH & ACADEMIC APPOINTMENTS

Doctoral Researcher — HEC Montréal / GERAD

2014–2018

Nonparametric (distribution-free) inference, prediction intervals, and high-dimensional and robust variable selection; ensemble methods for finite-mixture models.

Research Data Analyst — Tech3Lab, HEC Montréal

2016–2017

Longitudinal models and design of experiments for engagement measurement; custom expectation–maximization (finite-mixture) model with mixed effects for EEG data from a heterogeneous population.

Research Assistant — HEC Montréal

2010–2011

Drafting and dissemination of scientific work for academic and industry audiences (business-intelligence research group).

06 UNIVERSITY TEACHING

Lecturer, Statistics — HEC Montréal (2013–2014). Instructor of record for an undergraduate probability and statistics course, delivered to three sections (in French).

Graduate Teaching Assistant, Statistical Programming — HEC Montréal (2015). Graduate-level course in SAS and R statistical programming.

Teaching Intern, Undergraduate Mathematics — HEC Montréal (2010–2011). Tutorial instruction for first-cycle mathematics.

07 EDUCATION & DISTINCTIONS

Ph.D., Decision Sciences (Quantitative Methods / Computational Statistics) — Université de Montréal (program administered by HEC Montréal), 2019.

NSERC Alexander Graham Bell Canada Graduate Scholarship (CGS-D), Doctoral. Dissertation: *Three Essays on Nonparametric Prediction Intervals and Robust Variable Selection* (supervisor: Prof. Denis Larocque). External examiner: Prof. Hemant Ishwaran (University of Miami).

M.Sc., Business Intelligence (Statistics & Data Mining) — HEC Montréal, 2011.

With Great Distinction (GPA 4.09 / 4.3); Honour Roll (highest cumulative GPA, 2011). Thesis: *Robustness of Random Forests for Regression*.

B.B.A., Financial Markets — HEC Montréal, 2008.

Trilingual cohort (French / English / Spanish); exchange, Singapore Management University.

08 RESEARCH QUALIFICATIONS

Dr. Hoffmann's peer-reviewed research concerns whether a model's outputs deliver the reliability they claim — directly relevant to reconstructing, stress-testing, and critiquing statistical analyses. Her doctoral and master's work:

- Developed and validated new methods for prediction intervals with random forests, and evaluated whether nominal confidence levels achieve their actual empirical coverage.
- Studied estimator degradation under data contamination and anomalous observations, and developed robust alternatives to standard random forests.
- Modeled heterogeneous populations using finite-mixture and growth-mixture methods.
- Developed high-dimensional variable-selection and screening methods for settings with thousands of covariates ($p \gg n$), where conventional methods break down.
- Evaluated methods through controlled simulation and real-data analysis, and implemented algorithmic modifications at the source-code level (C / FORTRAN) integrated with established R software (randomForestSRC).
- The prediction-interval methods were subsequently implemented in a published CRAN R package (RFpredInterval) and featured in the *R Journal*.

09 PEER-REVIEWED PUBLICATIONS

Published under the author's former name, Marie H. Roy / Marie-Hélène Roy.

Roy, M.-H., & Larocque, D. (2020). Prediction intervals with random forests. *Statistical Methods in Medical Research*, 29(1), 205–229. https://doi.org/10.1177/0962280219829885 (First published online 2019.)

Roy, M.-H., & Larocque, D. (2012). Robustness of random forests for regression. *Journal of Nonparametric Statistics*, 24(4), 993–1006. https://doi.org/10.1080/10485252.2012.715161

Owen, V. E., Roy, M.-H., Thai, K. P., Burnett, V., Jacobs, D., Keylor, E., & Baker, R. S. (2019). Detecting wheel-spinning and productive persistence in educational games. *Proceedings of the Educational Data Mining (EDM) Conference*.

10 SELECTED PRESENTATIONS

Shapiro, S., & Roy, M.-H. (2024). Bridging the gap of MMM and MTA in a cookieless world. *I-COM Global Summit*, Málaga, Spain. (Co-chaired roundtable.)

Roy, M.-H., Larocque, D., & Dupuis, D. (2015). Robust variable selection with a multiple-step bootstrap procedure. *Joint Statistical Meetings (JSM)*, Seattle.

Roy, M.-H. (2013). A study of random forests using robust aggregation methods and splitting criterion. *Joint Statistical Meetings (JSM)*, Montréal.

Roy, M.-H. (2012). Robustness of random forests for regression. *International Conference on Robust Statistics (ICORS)*, Burlington.

Roy, M.-H. (2018). Adapting ensemble predictive modeling for educational video games. *IDEAS SoCal AI & Data Science Conference*, Los Angeles.

11 METHODS, COMPUTING & LANGUAGES

Causal & experimental methods. Incrementality and lift studies, geo experiments, difference-in-differences, A/B and holdout testing, design of experiments, propensity-score and quasi-experimental methods, treatment-versus-control validity.

Econometric & marketing-measurement methods. Marketing mix modeling (MMM), multi-touch attribution (MTA), market-response and price-elasticity (demand) modeling, media ROI, customer lifetime value, and time-series forecasting.

Uncertainty, robustness & inference. Nonparametric and distribution-free inference, prediction intervals and calibration, high-dimensional and robust variable selection, finite-mixture and growth-mixture models, and sensitivity and stability analysis.

Model diagnostics, validation & reproducibility. Model specification and diagnostics, cross-validation and benchmarking, simulation-based stress testing, drift monitoring, transformation and implementation review, and data and instrumentation integrity.

Predictive & machine-learning methods. Gradient boosting, random forests and ensembles, hierarchical and mixed-effects models, Bayesian models, clustering and segmentation, rare-event and anomaly detection, and IRT psychometrics.

Statistical computing & languages. R (primary), Python, SQL, SAS, C++ / FORTRAN; reproducible reporting (R Markdown, LaTeX). French (native), English (fluent / professional), Spanish (intermediate).